



VIRGINIA COMMONWEALTH UNIVERSITY

Statistical Analysis and Modelling (SCMA 632)

A3: Regression Analysis and Diagnostics

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Date of Submission: 25-06-2025

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Introduction

This report details a statistical analysis project conducted using R, encompassing two distinct datasets: the National Sample Survey Office (NSSO) 68th round data and Indian Premier League (IPL) cricket performance and salary data. The first part of the analysis focuses on understanding factors influencing household food consumption in a specific region using Ordinary Least Squares (OLS) regression. The second, more extensive part, delves into the world of IPL cricket to analyze player performance, its correlation with player salaries, and the underlying statistical distributions of individual player metrics. This comprehensive approach aims to demonstrate data handling, statistical modeling, and interpretative skills within the R environment.

Objectives

- **NSSO68 Dataset Analysis:**
 - Load, preprocess, and impute missing/infinite values for the "MEG" state dataset.
 - Perform OLS regression to identify key factors influencing *foodtotal_q*.
 - Assess multicollinearity using Variance Inflation Factors (VIF).
- **IPL Player Performance and Salary Analysis:**
 - Import and preprocess IPL performance and salary data.
 - Aggregate player performance (runs, wickets) by year.
 - Identify top 3 batsmen and bowlers for each IPL season.
 - Link player names across datasets using fuzzy matching.
 - Calculate the correlation between 2024 player salaries and total performance points.
 - Fit probability distributions (Normal, Gamma, Exponential, Log-Normal) to N. Pooran's runs per match.
 - Determine best-fitting distributions for top 3 batsmen's and bowlers' per-match performance from the last three seasons.
 - Visualize the linear regression model for salary vs. total points.

Business Significance

For NSSO68 Analysis:

The analysis of the NSSO68 dataset is crucial for understanding household consumption patterns. By identifying significant factors affecting *foodtotal_q*, policymakers can gain insights into the economic well-being of households. This information can inform targeted welfare programs, food security initiatives, and economic planning, helping to address disparities in consumption and improve living standards.

For IPL Analysis:

The IPL performance and salary analysis offers significant insights for various stakeholders:

- **Player Selection & Team Building:** Understanding performance trends and identifying top players across seasons allows team management to make data-driven

decisions during player auctions and squad formation, minimizing financial risk and maximizing team potential.

- **Salary Justification & Contract Negotiation:** By quantifying the relationship between player performance and salary, team management can better justify current player remuneration or structure future contracts based on on-field value, ensuring equitable compensation.
- **Strategic Planning:** Knowing the statistical patterns of player performance (e.g., the distribution of runs or wickets per match) can aid in game strategy, allowing coaches to anticipate player contributions and plan accordingly. This can also be valuable for fantasy sports analytics and sports betting.
- **Talent Identification:** By analyzing long-term performance and fitting distributions, teams can identify consistent performers and emerging talents, allowing for early investment in players who offer long-term value.

Results

Part (a): NSSO68 Dataset Analysis

The OLS regression model for *foodtotal_q* in the "MEG" state was fitted using *MPCE_MRP*, *Age*, *Meals_At_Home*, *Possess_ration_card*, *Education*, and *No_of_Meals_per_day* as predictors.

The summary of the regression revealed that:

- *MPCE_MRP* (Monthly Per Capita Expenditure) was a highly significant predictor, indicating that higher expenditure generally correlates with higher food consumption.
- *Age* and *Education* also showed significant relationships with food consumption, suggesting demographic and educational influences.
- *Meals_At_Home* and *No_of_Meals_per_day* were significant, as expected, directly reflecting consumption habits.
- *Possess_ration_card*'s impact was also observed, indicating the role of social support systems.
- The Variance Inflation Factors (VIF) were calculated to check for multicollinearity among the predictors. The VIF values were generally low (e.g., 1.0 to 1.5 for most continuous variables), suggesting that multicollinearity is not a significant issue in this model, meaning the predictors are largely independent of each other.

Part (b): IPL Player Performance and Salary Analysis

- **Top Players Each Season:**
 - The analysis successfully identified the top 3 batsmen (by total runs) and top 3 bowlers (by total wickets) for each IPL season. For the most recent 2024 season, the top performers identified were:
 - **Top 3 Batsmen (Total Runs - 2024):**
 - RD Gaikwad
 - V Kohli
 - B Sai Sudharsan

- **Top 3 Bowlers (Total Wickets - 2024):**

- HV Patel
- Mukesh Kumar
- Arshdeep Singh

How Salary Connects to Performance (2024 Season):

We matched player names between the performance and salary lists and created an overall "Total Points" score (runs + 25 times wickets).

Here's how salary connected to performance for 2024:

- **Salary vs. Runs (Batsmen):** A correlation of **0.453**. This is a moderate positive link, meaning more runs tend to go with higher salaries.
- **Salary vs. Wickets (Bowlers):** A correlation of **0.214**. This is a weaker positive link.
- **Salary vs. Total Points:** A correlation of **0.338**. This is also a moderate positive link.

Our simple prediction model (linear regression) for salary based on Total Points had an R-squared of **0.094**. This means only about 9.4% of the salary differences can be explained by our "Total Points" score alone. The average prediction error (Mean Squared Error) was **261,135.31**. The graph shows how our predicted salaries compare to actual salaries.

Best Statistical Patterns for Player Performance:

- For Batsman N. Pooran (runs per match):
When looking at N. Pooran's runs per match, the Gamma distribution was the best fit. This kind of pattern often describes data that starts at zero and then has a longer tail towards higher numbers, which makes sense for runs scored. The statistical test (Kolmogorov-Smirnov) showed a p-value of 0.03335 for this fit.
- For Other Top 3 Players (Last 3 Seasons):
We also found the best-fitting patterns for the top players in the last three seasons.
 - **Year: 2024**
 - **Top 3 Batsmen:** RD Gaikwad (Exponential), V Kohli (Exponential), B Sai Sudharsan (Gamma)
 - **Top 3 Bowlers:** HV Patel (Log-Normal), Mukesh Kumar (Log-Normal), Arshdeep Singh (Log-Normal)
 - **Year: 2023**
 - **Top 3 Batsmen:** Shubman Gill (Gamma), F du Plessis (Normal), DP Conway (Exponential)
 - **Top 3 Bowlers:** MM Sharma (Gamma), Mohammed Shami (Log-Normal), Rashid Khan (Log-Normal)
 - **Year: 2022**
 - **Top 3 Batsmen:** JC Buttler (Exponential), KL Rahul (Exponential), Q de Kock (Exponential)
 - **Top 3 Bowlers:** YS Chahal (Log-Normal), PWH de Silva (Log-Normal), K Rabada (Log-Normal)

Interpretations

For household spending, our findings confirm that money available, age, and education are strong indicators of how much food a family consumes. The low VIF values mean our chosen factors are distinct and not just measuring the same thing.

In cricket, the top players clearly show consistent high performance. This means they're very valuable to their teams.

The link between salary and performance is positive but not super strong. This tells us that while good performance leads to better pay, it's not the *only* thing teams consider. Other factors like how fast they score, how well they field, their leadership skills, or even their popularity, play a big part. For bowlers, who often take zero wickets in a match, their value might be more about keeping runs down or taking wickets at crucial moments.

The different statistical patterns we found (like Gamma for runs, Log-Normal for wickets) show that different aspects of cricket performance behave in unique ways. Runs often have a certain "skew" (more lower scores, fewer very high ones), while wickets often have many zero values and then some higher counts. The fact that our R code successfully found good fits for bowlers now means it's better at handling this kind of data with many zeros.

Recommendations

- **For Household Spending Data:**
 - Look deeper into how different factors combine. For example, how does age *and* education together affect spending?
 - Try different statistical methods that might give even more detailed insights into spending across various income levels.
- **For IPL Data:**
 - **Better Player Matching:** Improve how we match player names, maybe using unique player IDs if they become available, to ensure we link salaries to the correct performance records every time.
 - **More Performance Details:** Include more specific stats like strike rate, economy rate, and fielding metrics, as these also determine a player's worth.
 - **Smarter Salary Predictions:** Use more advanced statistical models that can capture the many complex reasons behind a player's salary, since simple performance doesn't explain everything.
 - **Categorize Players:** Analyze batsmen, bowlers, and all-rounders separately, as their value is judged differently.
 - **Track Over Time:** Study how player performance and salaries change throughout their careers, not just in one season. This helps understand their long-term value.
 - **Improve Zero-Heavy Data Analysis:** For things like wickets (where many players get zero in a game), use special statistical patterns designed for "count data" to get more accurate results.
 - **Check Our Work:** Always test our predictions and models thoroughly with new data to make sure they are accurate and reliable in real-world situations.

References

- Data provided through Canvas Portal.